

Deterministic Approximation and Share Inversion for Factor Probit with Many Alternatives

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Abstract

Multinomial probit choice probabilities are high-dimensional Gaussian integrals with no closed form; the standard tool, the GHK simulator, computes them one alternative at a time by smoothed importance sampling. For the factor-structured case $\Sigma = VV^\top + \text{diag}(D)$ we describe a reproducible approximation that computes *all* N probabilities together: conditionally on k Gaussian factors the alternatives are independent, and a lattice survival-product identity yields the entire share vector in $O(QNL)$ for Q factor nodes and L lattice points — Gauss–Hermite quadrature at low factor rank, fixed-design RQMC at higher rank. In tested $k = 2$ problems the maximum observed discrepancy from simulation references stayed below 10^{-3} from $N = 5$ to $N = 5000$ (21 seconds at $N = 5000$); a direct utility simulator at matched wall time reaches errors $1.6\text{--}6\times$ larger at these budgets and separates decisively below the 10^{-3} level, so the case for the method is error decay, reproducibility, resampling-free derivatives, and calibration — not raw forward speed alone. The share map is the gradient of the social-surplus function; its Jacobian is minus a weighted graph Laplacian, giving symmetric positive-definite reduced systems, $O(QNL)$ Jacobian–vector products, and a globally well-posed share inversion (existence, uniqueness modulo location) solved at $N = 1000$ in under a minute, with all N single-removal counterfactuals in $O(QN^2L)$ from one conditional field. The method requires the factor form: where Σ has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space, simulation at exact Σ remains the tool.

1 Introduction

When a consumer chooses one product from a large assortment, a commuter one route from a network, or a platform ranks one item above thousands of others, the modeling question is always the same: if an alternative disappears or changes, *where does its demand go?* Getting that right requires correlated utilities. The canonical model that allows them — multinomial probit (MNP), with jointly normal utilities and unrestricted substitution patterns — has been effectively locked away from exactly the settings that need it most. Its probabilities are $(N - 1)$ -dimensional Gaussian integrals with no closed form, and for three decades the field’s tool has

*Numbers in this paper are produced by committed, seeded scripts with correctness anchors run before any comparison (<https://github.com/microprediction/kinetics>, experiments 13–17; the exact commit is recorded in the repository README and an immutable tag is pinned at circulation). The algorithm ships in the **thurstone** package; package–benchmark agreement is itself a reported number. Benchmarks: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0; experiment 17 enforces single-threaded BLAS via **threadpoolctl** and reports median-of-3 timings for sub-second calls; long simulations are single realizations, disclosed as such. Monte Carlo chunk sizes derive from an explicit memory budget (1.5 GB); peak resident memory stays below 2 GB.

been the GHK simulator [1, 2, 3]: smoothed importance sampling, one alternative at a time, with $R^{-1/2}$ error in the number of draws. At N in the hundreds or thousands, that is a separate simulation per alternative per likelihood evaluation. So practice at scale retreats to the multinomial logit family — not because anyone believes proportional (IIA) substitution, but because logit is what remains computable. The cost of that retreat is measurable: in the substitution study of Section 3, plain logit misallocates about 23% of the demand released by a deleted alternative where the correctly structured probit model misallocates 5%.

This paper’s point is that for the covariance structure applied work most often actually has — a factor model, $\Sigma = VV^\top + \text{diag}(D)$ — the lock is not real. We give one algorithm that solves *both directions* of the probit share map, plus its derivatives and counterfactuals:

1. **Forward pass** (utilities $\rightarrow$ shares): all N choice probabilities at once in $O(QNL)$ for Q factor nodes and L lattice points. All 5000 shares of an $N = 5000$ problem in 21 seconds, at accuracy near the 10^{-4} noise floor of the simulation reference; matching that with our per-alternative GHK implementation extrapolates to roughly forty hours.
2. **Calibration** (observed shares $\rightarrow$ utilities): the probit analogue of the logit-side inversions used throughout demand estimation [10, 11]. Globally well-posed (Theorem 1: existence and uniqueness modulo location) and solved for $N = 1000$ in under a minute.
3. **Derivatives**: Jacobian–vector products in $O(QNL)$ with an exact weighted-graph-Laplacian structure — symmetric positive-definite reduced systems, hence Krylov methods — with no resampling noise.
4. **Counterfactuals**: all N single-removal share vectors from one conditional field pass in $O(QN^2L)$.

The mechanism is a single trick used four ways: conditionally on the k factors the alternatives are independent, and a lattice survival-product identity computes every alternative’s integral in one shared $O(NL)$ pass — the field’s survival product is built once and each alternative’s leave-one-out integrand is recovered by division.

Scope, honestly. The method needs the factor form: where Σ has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space, simulation at exact Σ remains the tool (Section 3). And at coarse accuracy targets ($\sim 10^{-3}$), plain direct utility simulation is a genuinely competitive forward baseline (Section 2); the decisive advantages are error decay below that level, reproducibility, resampling-free derivatives, and the inversion and counterfactual machinery. The one-pass identity itself comes from an unlikely literature — inferring ability from win probabilities in multi-entrant races [9] — which is perhaps why it has not been assembled for probit before, despite each ingredient being classical.

This paper concerns the factor-structured case

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i} \varepsilon_i, \quad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0, 1) \text{ i.i.d.}, \quad D_i > 0, \quad (1)$$

i.e. $\Sigma = VV^\top + \text{diag}(D)$. Factor dimension reduction for probit is itself classical — one-factor quadrature goes back to Butler and Moffitt [4], factor-analytic probit to Elrod and Keane [5] — and GHK is accurate and well understood at moderate N . The contribution here is the all-alternatives assembly, its inverse transform, the derivative and counterfactual structure, and benchmark plus boundary evidence for all of it.

Proposition 1 (Conditional identity). *Under (1), with $g_i(\cdot | f)$ and $F_i(\cdot | f)$ the conditional density and CDF of U_i given f ,*

$$p_i = \mathbb{E}_f \left[\int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx \right], \quad \sum_i p_i = 1, \quad (2)$$

and the product over $j \neq i$, for all i simultaneously, is obtained from $\log F_{\text{field}} = \sum_j \log F_j$ by subtraction.

Proof. Conditional on f the U_j are independent, so $\mathbb{P}(U_i \text{ maximal} | f) = \int g_i \prod_{j \neq i} F_j dx$; average over f . The events partition (ties are null for continuous laws), giving $\sum_i p_i = 1$. $\square$

Proposition 2 (Complexity). *With Q factor nodes and L lattice points, the forward share vector costs $O(QNL)$; the full single-removal ensemble $\{P(j \text{ chosen} | i \text{ removed})\}_{i,j}$ costs $O(QN^2L)$ arithmetic and $O(N^2)$ storage, reusing one conditional field construction.*

Proof. Per node: the field log-product costs $O(NL)$, each alternative's integral $O(L)$ after subtraction, giving $O(NL)$; the removal ensemble evaluates N survivor integrals for each of N removals, $O(N^2L)$, over the same cached field. $\square$

Proposition 3 (Identification and differentiability). *$p(\mu + c\mathbf{1}) = p(\mu)$, so $J = \partial p / \partial \mu$ has $\mathbf{1}$ in its null space and μ is identified only up to location. With B an $N \times (N - 1)$ basis of $\mathbf{1}^\perp$ and $\mu = B\eta$, share inversion and implicit differentiation are taken in reduced coordinates: $d\eta/d\theta = -(B^\top JB)^{-1} B^\top \partial p / \partial \theta$, provided $B^\top JB$ is nonsingular.*

Proof. Invariance holds because a common location shift does not change the argmax; differentiating $p(\mu + c\mathbf{1}) = p(\mu)$ in c gives $J\mathbf{1} = 0$. Differentiating $p(B\eta(\theta), \theta) = \hat{p}$ in θ and projecting with $B^\top$ gives the reduced formula under the stated regularity. $\square$

Theorem 1 (Weighted-Laplacian structure and global inversion). *For model (1) with $D_i > 0$, in max-wins coordinates, define for $i \neq j$*

$$w_{ij} = \mathbb{E}_f \int g_i(x | f) g_j(x | f) \prod_{\ell \neq i,j} F_\ell(x | f) dx > 0.$$

Then $J = \partial p / \partial \mu$ satisfies $J_{ij} = -w_{ij}$ and $J_{ii} = \sum_{j \neq i} w_{ij}$, i.e.

$$J = \sum_{i < j} w_{ij} (e_i - e_j)(e_i - e_j)^\top, \quad h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2,$$

a weighted graph Laplacian with everywhere-positive weights: J is symmetric, positive semidefinite, has $\mathbf{1}$ as its only null direction, and $B^\top JB \succ 0$ for every full-rank basis B of $\mathbf{1}^\perp$. Moreover $p = \nabla G$ for the social-surplus function $G(\mu) = \mathbb{E} \max_i \{\mu_i + \xi_i\}$, and for every strictly positive target $\hat{p}$ the inverse problem $p(\mu) = \hat{p}$ has exactly one solution on $\mathbf{1}^\perp$.

Proof. Differentiating (2) in μ_j and integrating by parts gives $J_{ij} = -w_{ij}$ for $i \neq j$; the diagonal follows from $J\mathbf{1} = 0$ (Proposition 3), and positivity of w_{ij} from positivity of Gaussian densities. The quadratic form is then immediate, and it vanishes only for constant h . For inversion,

minimize $H(\mu) = G(\mu) - \hat{p}^\top \mu$ over $\mathbf{1}^\perp$: H is strictly convex there (its Hessian is the reduced J), and since the shocks have zero mean, $G(\mu) \geq \max_i \mu_i$, so

$$H(\mu) \geq \max_i \mu_i - \hat{p}^\top \mu = \sum_i \hat{p}_i (\max_j \mu_j - \mu_i) \geq \hat{p}_{\min} (\max_i \mu_i - \min_i \mu_i) \rightarrow \infty$$

as $\|\mu\| \rightarrow \infty$ on $\mathbf{1}^\perp$. The unique minimizer satisfies $\nabla G = \hat{p}$. (Every identity here is verified numerically in the repository: the w_{ij} representation to 6×10^{-9} against finite differences, symmetry to 4×10^{-12} , and positive definiteness of the reduced numerical Jacobian; the theorem statement was proposed by a referee.) Social-surplus convex duality for discrete choice is classical [14, 15]; the contribution here is not well-posedness itself but its scalable realization. $\square$

Proposition 4 (Jacobian–vector products in $O(QNL)$). *With hazards $\lambda_j = g_j/F_j$, $\Lambda = \sum_j \lambda_j$, and $A = \sum_j h_j \lambda_j$ (all conditional on f , all functions of x), the Jacobian of the exact max-wins map acts on a direction h as*

$$(Jh)_i = \mathbb{E}_f \int g_i \left(\prod_{j \neq i} F_j \right) (h_i \Lambda - A) dx,$$

where the field sums Λ and A are formed once per node and lattice point, so all N components cost $O(NL)$ per node. Translation invariance is visible ($h = \mathbf{1}$ gives $A = \Lambda$, hence $Jh = 0$), and the weighted-Laplacian form of Theorem 1 follows by expanding A . This enables Krylov solves of the reduced system in Proposition 3 without forming J densely at $O(QN^2L)$.

Proof. Differentiate (2): the μ_i -derivative of g_i contributes $\int g'_i \prod_{j \neq i} F_j$, which after integration by parts equals $\int g_i \sum_{j \neq i} \lambda_j F_j^{-1} \cdot (\prod_{\ell \neq i} F_\ell) g_j / \lambda_j$ -terms assembled as $\int g_i \prod_{j \neq i} F_j (\Lambda - \lambda_i)$; the μ_j -derivatives of the F_j factors contribute $-w_{ij} h_j$ summed to $-\int g_i \prod_{j \neq i} F_j (A - h_i \lambda_i)$. Combining gives the display. $\square$

Which map is differentiated, exactly. Three maps must be distinguished: the exact max-wins map $p^{\max}(\mu)$; the reflected min-wins map $p^{\min}(a)$ used by the implementation, related by $a = -\mu$ so that $D_\mu p^{\max}[h] = D_a p^{\min}[-h]$; and the implemented approximation $p_{Q,L} = \tilde{p}_{Q,L} / (\mathbf{1}^\top \tilde{p}_{Q,L})$, whose derivative carries a normalization correction: with $v = D\tilde{p}[h]$ and $s = \mathbf{1}^\top \tilde{p}$,

$$Dp_{Q,L}[h] = \frac{v - p_{Q,L} (\mathbf{1}^\top v)}{s}.$$

Since $\mathbf{1}^\top v = 0$ analytically, the correction is of the order of the quadrature defect; empirically the uncorrected formula matches central finite differences of the returned normalized map to 5×10^{-9} . One further convention: the adaptive lattice endpoints depend on μ , so derivative statements refer to the *fixed-design* map — the lattice envelope and factor nodes are frozen during any inversion or outer-derivative computation, and the inversion’s analytic own-location slopes are accordingly a Jacobi quasi-Newton preconditioner rather than the diagonal of the adaptive map’s Jacobian.

Tail-stable implementation. A referee exhibited a modest problem ($N = 8$, heterogeneous D) on which our published derivative code returned NaN/infinity: the density was floored before its logarithm while the exact log-survival stayed finite, producing spurious e^{+92} hazards. The implementation now works in the log domain throughout — $\log \Phi$ directly (never $1 - \Phi$, which

underflows at $z \approx 8.3$), analytic $\log g$, hazards as $\exp(\log g - \log \Phi)$ (bounded, $\sim z$ in the tail) — and the failing case is a regression test. The same fix applies to the forward map, where deep-tail shares now resolve to genuine small positives rather than floored zeros.

The factors need not be Gaussian: (2) requires only conditional independence and an integrable factor law, so Student, mixture, or empirical factor distributions are admissible with appropriate nodes, and the base densities may be arbitrary and alternative-specific. One caution for asymmetric laws: the implementation works in the reflected (min-wins) race, and $-U$ has the same distribution as the reflected model only when the factor and base laws are symmetric; for asymmetric laws the reflected distributions must be used explicitly.

Numerical method, stated completely. The benchmarks use $L = 1501$ lattice points on the adaptive interval $[\min_{i,q} m_i(f_q) - 8 \max_i \sqrt{D_i}, \max_{i,q} m_i(f_q) + 8 \max_i \sqrt{D_i}]$, where $q = 1, \dots, Q$ indexes the retained factor nodes. Factor integration: the GHK benchmark (experiment 13) uses product Gauss–Hermite of order 15/11/9 for $k = 2/3/4$; the boundary experiment (experiment 14) uses order 15 per dimension for all $k \leq 4$, which after pruning weights below 10^{-7} of the maximum leaves $Q = 145, 1317$, and 10929 nodes for $k = 2, 3, 4$ — making the exponential growth of Q with rank explicit; scrambled-Sobol nodes (2^{13} points, fixed seed) are used for $k > 4$: reproducible fixed-seed randomized QMC, disclosed as such. Survival floors sit at 10^{-300} and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2); statements about derivatives refer to derivatives of this approximation, whose fidelity to exact MNP derivatives is governed by L and Q .

Inversion. Given target shares $\hat{p}$ and (V, D) , the inverse transform finds mean-zero μ with $p(\mu) = \hat{p}$ by a damped Jacobi-style diagonal-Newton iteration: all coordinates updated simultaneously using own-location slopes of the unnormalized lattice map (available from the same pass), the field refrozen each iteration, a warm start from the independent-field inverse using total variances, and a tail-aware tolerance (alternatives with target shares below $10^{-4}/N$ are matched best-effort; their locations are weakly identified — itself a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear). Convergence in a handful of iterations is an empirical observation on the tested problems, not a theorem. The design follows the original transform’s inversion [9] and the calibration practice of [17].

2 Benchmark against GHK

Anchor first. On the binary case, where MNP has a closed form, the lattice transform agrees to 3.3×10^{-16} and our GHK implementation — validated before being compared against — to 2×10^{-14} at $R = 2 \times 10^5$ (an easy anchor for both methods, useful for code validation rather than as evidence about the hard regime). At $N = 5$ both methods sit within the noise of a 10^7 -draw Monte Carlo reference, and the shipped `thurstone.FactorRace` agrees with the benchmark kernel to 1.1×10^{-5} .

Error metric: throughout, $\max_i |\hat{p}_i - p_i|$ against a simulation reference. A maximum over N coordinates tends to grow with N at fixed per-coordinate accuracy, and the reference’s own maximum-coordinate noise behaves likewise; mean-absolute and total-variation summaries are committed alongside (experiment 16): at $N = 1000$ the lattice’s mean absolute error is 5.6×10^{-6} and total variation 2.8×10^{-3} against a fresh 2×10^6 -draw reference.

N	lattice transform	GHK ($R = 1000$)
5	20 ms, err 5.7×10^{-4}	1 ms, err 3.1×10^{-3}
20	87 ms, err 4.4×10^{-4}	12 ms, err 4.4×10^{-3}
50	214 ms, err 2.7×10^{-4}	75 ms, err 7.9×10^{-3}
200	827 ms, err 3.0×10^{-4}	1275 ms, err 6.8×10^{-3}
1000	4.1 s, err $6.8 \times 10^{-4} \dagger$	—
5000	20.7 s, err $5.0 \times 10^{-4} \dagger$	—

Table 1: Full share vector, $k = 2$ factors, one problem per size (utility spread 1.0 through $N = 200$, 1.5 above). Reference: 2×10^6 MC draws through $N = 200$; 5×10^5 draws at $N = 1000$ and 5000 ($\dagger$: there the reference’s own maximum-coordinate noise is of comparable order, so these entries bound the lattice error only to that order). A replication study at common spread 1.0 (experiment 16; ten problems per size at $N \leq 200$, three at $N = 1000$, twin independent 2×10^6 -draw references per problem) gives worst-case lattice error $3.5\text{--}4.6 \times 10^{-4}$ at every size, at or below the references’ own replicate-to-replicate noise. Lattice error stayed below 10^{-3} throughout; GHK cost crossed over by $N \approx 200$.

Matched-accuracy extrapolation (not a measurement). Fitting a power law to the reference implementation’s timings over $N \in [20, 200]$ ($\alpha \approx 2.0$; a lower bound, the asymptotic per-share cost being cubic) and combining with $R^{-1/2}$ error scaling suggests that matching the lattice’s 5×10^{-4} at $N = 5000$ would take this implementation on the order of forty hours (0.25 h at $R = 1000$, times $(6.8 \times 10^{-3}/5 \times 10^{-4})^2 \approx 185$), against 21 seconds. The baseline is a straightforward single-threaded Python GHK: per-alternative Cholesky, plain pseudorandom uniforms, one realization per problem, shares normalized by their sum. It establishes the scaling of the per-alternative simulation approach, not a field-wide impossibility. The strongest cheap baseline is *direct utility simulation* — draw utilities, take argmax, average — which produces the whole share vector per draw and, unlike GHK, does not pay per alternative. We ran it at wall time matched to the lattice call (experiment 16): its maximum-coordinate error was 1.6–6 $\times$ the lattice’s at every tested N (6.1×10^{-4} versus 3.9×10^{-4} at $N = 1000$, 0.5M draws in the matched 4.3 seconds). At the 10^{-3} accuracy level, then, raw forward speed does not by itself separate the methods. What does is error decay, and this is now measured rather than asserted (experiment 17, $N = 200$, $k = 2$): holding the factor rule fixed, lattice self-convergence is 2.7×10^{-8} at $L = 101$ and at machine precision from $L \approx 200$ — the integrand is analytic and endpoint-decaying, so the uniform lattice is effectively spectrally accurate and L -error is subdominant; holding L fixed, the Gauss–Hermite sweep decays from 5.3×10^{-4} (order 3) to 5.9×10^{-9} (order 15); the pre-normalization defect $|1 - \mathbf{1}^\top \tilde{p}|$ is 2.4×10^{-8} . On the same problem the measured accuracy–time frontier (Figure 1, left) has the lattice at 4.1×10^{-5} error in 0.03 seconds (at the twin-reference noise floor), direct MC at 1.2×10^{-4} in 15.7 seconds ($R = 10^7$), scrambled-Sobol direct QMC at 2.0×10^{-4} in 2.6 seconds (2^{20} points), and GHK at 2.0×10^{-3} in 10.1 seconds ($R = 10^4$): at accuracy targets below 10^{-3} the deterministic map separates decisively, consistent with $R^{-1/2}$ against a rapidly-converging quadrature. Remaining stronger baselines — QMC-GHK, minimax tilting [8] — are queued in the repository.

Derivatives, stated correctly. For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of μ whose derivatives are evaluated without

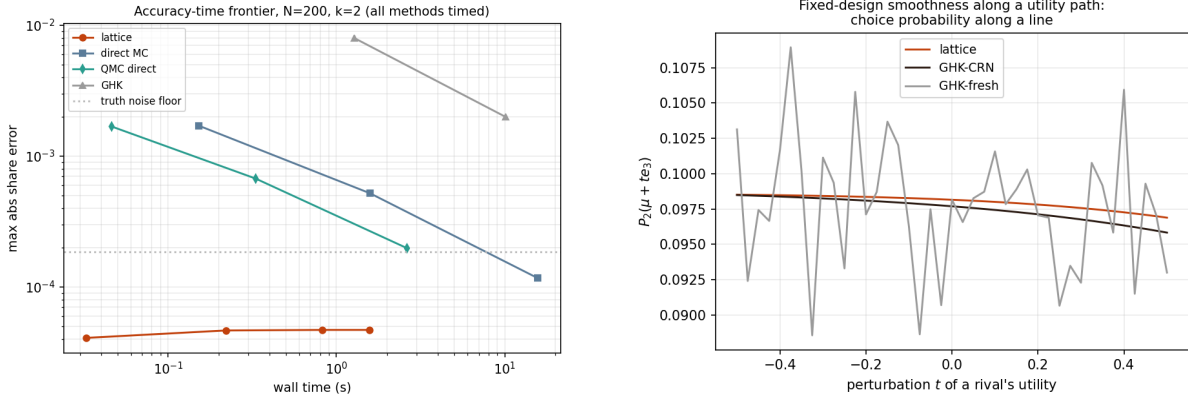


Figure 1: Left: measured accuracy–time frontier at $N = 200$, $k = 2$ (experiment 17) — every method timed on the same problem against twin 5×10^7 -draw references; direct simulation is a first-class baseline. Right: fixed-design smoothness along a utility path. Fresh-draw GHK shows run-to-run simulation noise (roughness statistic — maximum second difference over dt^2 — 39.3, versus 0.01 for both common-random-number GHK and the lattice); CRN-GHK is smooth conditional on its draw set, and the offset between its curve and the lattice’s is a discrepancy between two approximations whose decomposition requires an independent high-accuracy reference, queued. All three are approximations.

resampling; analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws [2, 3]. The practical distinction is the character and controllability of the approximation error — quadrature resolution versus draw count — not the possibility of differentiation.

Share inversion. At $N = 1000$, $k = 2$, with targets from 5×10^6 MC draws floored at 10^{-7} and renormalized: the inversion completes in 57 seconds. The iteration solves its own equations to 6.4×10^{-9} — a statement about the solver, not statistical accuracy, since the target carries sampling noise $\sim 2 \times 10^{-4}$ — and recovers the utilities of identified alternatives (target share above 3×10^{-4}) to maximum error 0.017. A replication study (experiment 16) repeats the inversion on a second $N = 1000$ problem against three independent 5×10^6 -draw targets: 57 seconds each, 161–162 alternatives identified (98.1% of share mass), recovery error maximum 0.015–0.023 and median ≈ 0.0027 across replicates, with the recovery-versus-share curve committed alongside. Numerical well-posedness matches Theorem 1: the reduced Jacobian assembled from N JVP calls at $N = 50$ is symmetric to 1.7×10^{-18} , has row-sum defect 6.2×10^{-17} , and reduced eigenvalues in $[2.5 \times 10^{-5}, 0.22]$ — positive definite as proved. We are not aware of probit share inversion at this scale in current simulation-based practice.

Removal ensemble. All 200 single-removal share vectors from one conditional field construction in $O(QN^2L)$: 51.5 seconds versus 172.4 seconds recomputing ($3.3\times$), with ten evenly spaced removals checked directly (maximum discrepancy 2.8×10^{-17}).

3 Boundary experiments

General covariance through the factor approximation. The method requires $\Sigma \approx VV^\top + D$; GHK does not. Choices depend on (μ, Σ) only through $P\mu$ and $P\Sigma P$, $P = I - \mathbf{1}\mathbf{1}^\top/N$: a common factor loading $V \rightarrow V + \mathbf{1}c^\top$ shifts every conditional location equally and cannot move the argmax (machine-precision fact, tested), so the factor fit must be run in this choice-relevant quotient space — fitting raw Σ can spend scarce rank on an irrelevant common component ($\Sigma = \tau^2\mathbf{1}\mathbf{1}^\top + \beta\beta^\top + D$ with large τ : the raw rank-1 fit takes $\mathbf{1}\mathbf{1}^\top$ and misses $\beta\beta^\top$ entirely; on such an example the contrast-space fit cuts rank-1 share error from 4.6×10^{-2} to 8.3×10^{-3}). A caution from a wrong first implementation: only the common *factor* direction is irrelevant; a common addition to D inflates every difference variance and is choice-relevant, so D must come from the quotient fit. Loadings are centered and SVD-canonicalized (ordered columns, fixed signs), making results reproducible at the covariance level. With this fit: one random correlation matrix at each spectral decay $\lambda_m \propto m^{-\gamma}$, $\gamma \in \{0.5, 1.5, 3\}$, at $N = 50$ (single realizations), share error against an 8×10^6 -draw reference, GHK at the exact Σ for comparison. The plotted quantity is the rank- k approximation error of the stated procedure, not a minimum over factor models. Findings: error decays fastest for near-low-rank truth ($\gamma = 3$: 4.3×10^{-2} at $k = 1$ to 1.5×10^{-3} at $k = 8$) and slowest at intermediate decay ($\gamma = 1.5$: 2.0×10^{-2} to 2.3×10^{-3}). In these three single-matrix illustrations the rank-8 approximation at the stated Q, L had lower observed maximum discrepancy than GHK at $R = 10^4$ (3.2 – 4.1×10^{-3}); this was *not* a wall-time-matched comparison — at this size the $k = 8$ RQMC evaluation takes ~ 14 s against 0.5–0.7 s for GHK at $R = 10^4$ (timings now in the committed output), so at $N = 50$ GHK is the cheaper route to 3 – 4×10^{-3} ; the lattice’s advantage is large N at small k , not this corner. Where accuracy demands exceed the achievable rank- k error, GHK’s arbitrary- Σ generality is the right tool.

Substitution fidelity: a 2×2 factorial. Why want probit-style flexibility at scale? An earlier version of this experiment could not attribute its increments: its Gumbel candidate received factor loadings *and* alternative-specific scales at once (making it a heteroskedastic factor-Gumbel race rather than mixed logit). The redesign isolates the axes. Ground truth is misspecified for every candidate — $t(5)$ factors and skew-normal idiosyncratic noise standardized to mean zero and unit variance, homoskedastic; the race is computed min-wins, so positive skew in race coordinates is *left*-skew in utility coordinates. The four candidates hold the idiosyncratic scale fixed at one and vary $\{\text{Gumbel, Gaussian}\}$ base $\times \{V = 0, V = V^*\}$: independent Luce (= plain multinomial logit exactly), independent probit, factor mixed logit (common-scale, the term now used correctly), and factor probit. All are calibrated to the same full-menu shares (residuals $\leq 4 \times 10^{-10}$). Deletion truths come from 2×10^7 *common* utility draws with the top three alternatives retained per draw, from which every single-deletion and pair-deletion truth follows with common random numbers: 105 informative blocks stratified by deleted mass (26 / 29 / 28 blocks at $>10\%$ / 2–10% / 0.5–2%), against 24 random blocks before. Mean misallocated fraction of redistributed mass:

	mass $> 10\%$	2–10%	0.5–2%
independent Luce (= plain logit)	0.229	0.245	0.261
independent probit	0.210	0.232	0.230
factor mixed logit	0.114	0.146	0.199
factor probit	0.045	0.062	0.094

The factorial decomposition (top two strata): the factor increment is dominant in both families (+0.115/+0.099 within Gumbel, +0.165/+0.170 within Gaussian); the family increment alone is small (+0.019/ +0.013 at $V = 0$) but grows with factors present (+0.069/ +0.084), a negative interaction ($-0.050/-0.071$): factor structure and the Gaussian base are complements in this design. Caveats: a single truth design and seed (replication across skewness, tails, and a separate heteroskedasticity axis is queued), and the skew-normal truth is Gaussian-adjacent. The figure shows individual block observations, not bars.

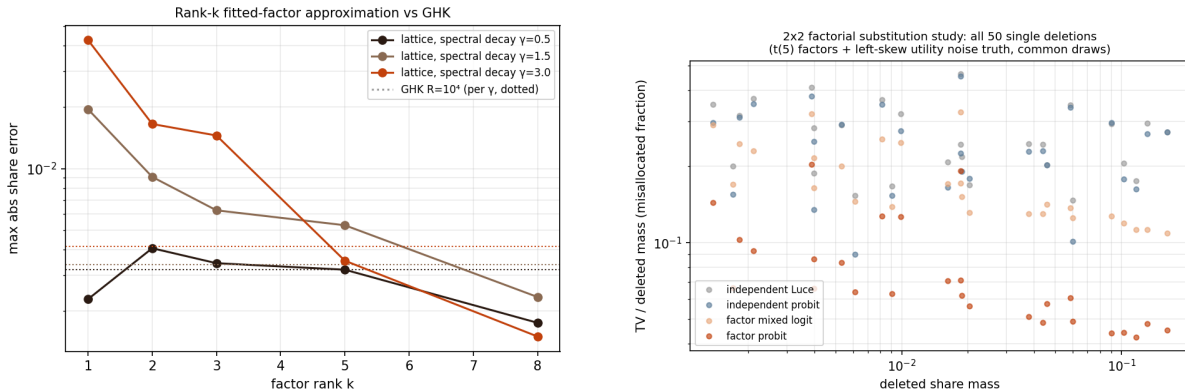


Figure 2: Left: rank- k contrast-space approximation error under spectral decay γ (dotted: GHK at $R = 10^4$, per γ ; not wall-time matched). Right: the 2×2 factorial substitution study — every informative single-deletion block shown individually.

4 Related work

GHK and simulated likelihood are standard [1, 2, 3]; analytic-approximation lines include MACML [6]; modern Gaussian-probability methods include minimax tilting [8]. Factor dimension reduction for probit is classical: Butler–Moffitt one-factor quadrature [4], Elrod–Keane factor-analytic probit [5], and the probit probability approximations surveyed in [7]. The lattice survival-product identity and its inverse transform are from [9]. On inversion, the social-surplus identity $p = \nabla G$ and convex-conjugate formulation of the inverse are established discrete-choice results — Chiong, Galichon and Shum [14] formulate the inverse through the convex conjugate, and Norets and Takahashi [15] establish surjectivity — and factor-structured MNP for large choice sets has been pursued via Bayesian estimation by Loaiza-Maya and Nibbling [16]. The novelty claimed here is accordingly *not* well-posedness of inversion, nor dimensional reduction, both old: it is the scalable deterministic realization of the forward and inverse maps — all-alternatives assembly in $O(QNL)$, the $O(QNL)$ Jacobian–vector product, and shared-field deletion counterfactuals in $O(QN^2L)$. Share inversion and its acceleration are central to demand estimation [10, 11]; automatic differentiation of logit-based BLP is recent [12]; a 2026 preprint approximates correlated-choice probabilities with equivariant neural networks [13], evidence of current demand for this computation. We are unaware of an automatically differentiable, large-choice-set factor-probit demand implementation built on deterministic share inversion; that precise object is what Section 5 proposes.

5 Discussion

Estimation is the natural next step: with the reduced coordinates of Proposition 3, gradients of an outer likelihood or GMM objective pass through the share-inversion fixed point by implicit differentiation, a candidate route to differentiable probit demand estimation at large N . Estimating (V, D) rather than supplying them raises the usual identification cautions: utility scale normalization, factor rotation, choice-invariant common loading components, and the fact that only difference covariances are identified. Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities; a *common-scale* Gumbel base with zero loadings reproduces the Luce model exactly (alternative-specific scales break Luce, as they should), yielding correlated-softmax generalizations. Finally, the inversion inherits the original transform’s interpolation structure (per-node location-to-probability curves are cross-correlations, evaluable at all offsets simultaneously), which we have not yet exploited.

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